

Graphical Abstract

patX: Interpretable Pattern Extraction for Time Series and Spatial Data

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Highlights

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- patX learns interpretable B-spline templates as explicit features.
- patX is best on 6 of 7 biomedical benchmarks under matched protocols.
- Ablations show sliding-window matching, transforms, and search budget are task-dependent.

patX: Interpretable Pattern Extraction for Time Series and Spatial Data

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Abstract

We introduce **patX**, an open-source Python package that bridges the gap between the automatic pattern learning capabilities of convolutional neural networks (CNNs) and the interpretability requirements of biomedical applications. Like CNNs, patX automatically discovers discriminative patterns from input data and uses them for prediction; however, unlike CNNs where learned patterns remain opaque, patX makes this pattern learning explicit and interpretable through human-readable B-spline templates. By framing pattern discovery as a constrained optimization problem over B-spline templates across multiple signal transformations (e.g. raw, cumulative sum, DCT, FFT power, wavelet), patX learns compact, highly predictive patterns that serve as transparent input features for conventional machine-learning models. patX also supports spatially or temporally invariant pattern detection. By default, it evaluates a learned pattern in a sliding window over an allowed range and uses the best match (minimum MSE) as the feature value. This is analogous to applying a convolutional filter across positions followed by global pooling, yielding translation invariance.

We evaluate patX on seven diverse biomedical tasks including glucose forecasting, arrhythmia classification and gene expression prediction. We compare against standard automated feature extraction baselines and a CNN baseline under matched splits and metrics.

Beyond predictive performance, patX’s interpretable pattern discovery can lead to novel scientific insights: from histone modification data, it can reveal which modifications are most informative at specific genomic positions; from ECG data, spectral features may uncover potential electrophysiological causes of arrhythmias; and from glucose data, different temporal patterns can reveal the relative importance of short-term versus long-term dynamics for different patient populations. This transparent approach to automatic

feature extraction addresses the “black box” criticism of deep learning while maintaining competitive performance, making patX particularly suitable for clinical applications where explainability is crucial. Our tool is available at <https://github.com/Prgrmmrjns/patX> and offers a simple interface for extracting interpretable patterns from time series and spatial data.

Keywords: interpretable machine learning, time series, pattern discovery, B-splines, feature extraction

1. Introduction

Biomedical machine learning increasingly operates on time series and structured signals: continuous bedside monitoring (ECG, EEG, oxygenation, glucose), longitudinal electronic health records, wearable inertial sensors, and spatially indexed molecular measurements such as epigenomic tracks. These modalities are central to both clinical decision support and biomedical discovery [1, 2, 3]. However, in many biomedical settings the bottleneck is not only predictive accuracy, but also whether a model produces evidence that is actionable and auditable by domain experts [4, 5].

Historically, three paradigms have dominated signal modeling. First, manual feature engineering uses domain knowledge to craft descriptors with clear meaning (e.g., ECG morphology, EEG band power), but this approach is labor-intensive and may miss task-specific patterns in heterogeneous cohorts [6, 7]. Second, automated feature extraction libraries compute large feature sets at scale (e.g., tsfresh, catch22), but they can introduce feature-selection burden and dilute interpretability in high-dimensional spaces [8, 9, 10]. Third, deep learning learns representations end-to-end and often excels on benchmark tasks, yet the learned features are frequently difficult to validate in biomedical workflows where transparency and error analysis are essential [7, 11, 12].

In parallel, a rich line of work has explored *pattern-based* time series methods. Shapelets and related discriminative subsequence approaches demonstrate that compact, human-interpretable patterns can be highly predictive [13, 14, 15, 16]. Similarly, motif discovery seeks repeated or predictive subsequences in long physiological recordings and has been applied to biomedical signals [17, 18, 19]. These approaches highlight an attractive direction for biomedical modeling: represent samples through similarity to a small number of patterns that can be inspected, visualized, and related to known physi-

ology. Yet in practice, pattern discovery must also handle multi-channel signals, misalignment, and the fact that discriminative evidence may appear in different signal representations (time, frequency, cumulative dynamics) depending on the task.

We present **patX**, an interpretable pattern extraction framework that combines pattern-based modeling with Bayesian optimization and Machine Learning. patX learns smooth spline patterns and converts each input sample into a small, transparent feature vector of pattern-match scores. patX first performs a *multi-representation* search: it precomputes a library of signal transformations and automatically selects the most informative representations before pattern discovery. patX then supports translation-invariant matching by evaluating a pattern in a sliding window over an allowed search interval and using the best match, implemented efficiently via min-MSE pooling. This provides robustness to temporal or spatial misalignment while retaining an explicit, human-readable pattern representation.

We evaluate patX across seven biomedical tasks spanning arrhythmia classification on MIT-BIH [20, 21], emotion classification from EEG (Emotions), gene expression prediction from histone modification profiles (REMC) [22], ARDS prediction from MIMIC-IV clinical time series [23], glucose forecasting (AZT1D), human activity recognition from IMU sensors (PAMAP2) [24], and voice pathology detection (SVD). We compare against established automated feature baselines (tsfresh, catch22) and a CNN baseline under matched splits and metrics.

Our contributions are: (i) a novel freely available open-source framework for pattern discovery in (biomedical) signals that is easy to implement, individualize and extended upon a users specific needs, (ii) an evaluation across heterogeneous tasks illustrating that a small number of patterns can achieve competitive performance, and (iii) a general-purpose framework that learns interpretable spline patterns as explicit features for biomedical signals opening up new opportunities for knowledge discovery in (biomedical) datasets.

2. Related Work

This article is structured as follows. We first review related work on automated feature extraction and interpretable pattern-based representations for time series and structured biomedical signals. We then describe the patX methodology, report empirical results, and discuss practical implications for biomedical use.

2.1. Manual feature engineering in biomedical studies

Before the recent surge of automated feature extraction and deep learning, biomedical time series modeling was often driven by domain expertise. Practitioners designed task-specific descriptors—for example, morphology- and variability-related summaries for physiological signals—and trained classical models on these engineered features. This approach remains common in biomedical practice because it offers clear semantics, supports targeted error analysis, and can be more data-efficient than end-to-end representation learning. In glucose prediction, for example, careful preprocessing and feature design around known physiological dynamics is often critical [25]. While effective, manual feature engineering can be labor-intensive and may miss subtle, task-specific patterns that vary across cohorts or measurement regimes [6, 7].

2.2. Feature engineering packages and pipelines

To situate patX among commonly used toolkits, Table 1 summarizes representative libraries and transforms used for biomedical time series feature extraction.

Table 1: High-level comparison of time series feature engineering toolkits and transforms commonly used in biomedical pipelines.

Tool	Feature type	Strengths	Limitations / notes
patX (this work)	Learned spline patterns + match scores	Compact, human-readable patterns; localization of best match; adapts via transform selection	Search cost depends on optimization budget; restricted by chosen transform and pattern families
CMDA [26]	Workflow + hand-crafted features	Convenient end-to-end workflow for importing, preprocessing, and feature extraction for monitoring data	Primarily library-driven descriptors; interpretability depends on selected features
tsfresh [8]	Large library of hand-crafted features	Broad coverage of statistical/spectral descriptors; strong baseline for many tasks	High-dimensional feature sets can require selection and reduce transparency
catch22 [9]	Small canonical feature set	Compact and efficient fixed feature set; useful when speed and simplicity are critical	May miss task-specific evidence that requires custom patterns or transforms
hctsa [27] / TSFEL [28]	Large feature libraries	Wide coverage of engineered descriptors for exploratory analysis	Interpretability and selection burden increase with feature count
ROCKET / MiniROCKET [29, 30]	Fast random convolutional feature transforms	Extremely fast, competitive accuracy with simple classifiers; good for large datasets	Features are not directly interpretable; produces high-dimensional features

2.3. Automated feature extraction for time series

One approach to time series modeling is to compute a large collection of hand-crafted descriptors and train a conventional learner on the resulting feature matrix. This strategy underlies widely used libraries such as tsfresh and catch22 [8, 9], as well as broader feature sets such as hctsa and TSFEL [27, 28]. For monitoring data, CMDA provides a modular workflow spanning importing, preprocessing, and feature extraction [26]. These packages are attractive baselines in biomedical pipelines due to their generality and ease of use, but they can trade interpretability for breadth, and high-dimensional feature sets often require feature selection and careful validation [10].

2.4. Fast generic feature transforms

Another line of work seeks transforms that yield strong accuracy with very low computational overhead. ROCKET and MiniROCKET transform time series using large numbers of random convolutional kernels and then fit a simple classifier, achieving competitive accuracy at high speed [29, 30]. While effective, these representations are typically high-dimensional and not directly interpretable, which can limit their use in settings where users need human-auditable evidence.

2.5. Shapelets and shapelet selection for feature engineering

Shapelets are discriminative subsequences that can serve both as features and as an interpretable explanation of class differences [13, 14, 15]. Subsequent work has emphasized learning or selecting compact shapelet sets to improve efficiency and interpretability. Fast discovery schemes reduce the cost of enumerating candidates [31], while optimization-based approaches learn shapelets or their positions more efficiently [16, 32]. Fang et al. propose an efficient scheme to learn interpretable shapelets by operating in a compressed candidate space and refining shapelets with a classifier [33].

Most closely related to patX, Chen and Wan propose localized shapelet selection to incorporate location information into the selection and transform process, improving interpretability by explicitly modeling where a shapelet occurs [34]. patX differs in three key ways. First, we optimize patterns *end-to-end* using an objective tied to the validation score of the downstream machine learning model trained on pattern match features, rather than scoring candidates with a separate distance or location heuristic. Second, patX searches over multiple signal representations via a transform library and automatic transform selection, allowing patterns to emerge in the domain most predictive for a given biomedical task. Third, patX supports position invariance through sliding-window min-MSE pooling over a user-defined search interval, while still returning the best-match location for inspection. Finally, patX discovers patterns iteratively: it adds one pattern at a time and stops when adding a new pattern no longer improves validation performance. This greedy construction tends to yield compact feature sets in which each pattern provides measurable predictive value, reducing unnecessary features.

2.6. Interpretable modeling in high-stakes biomedical settings

In biomedical applications, interpretability is not solely a property of an explanation method, but a constraint that shapes what models can be

deployed and trusted. Clinician-centered studies and high-stakes decision-making arguments emphasize the need for representations that support error analysis and domain validation [4, 5, 35]. patX contributes to this direction by producing explicit patterns as features and by offering flexible extension points (transforms, distance metrics, and learners) to adapt the feature extraction process to domain constraints.

3. Methods

3.1. Overview

patX discovers interpretable patterns from time series and spatial data through a three-stage pipeline: (1) signal transformation, (2) transform selection, and (3) iterative pattern discovery. The core idea is simple: learn spline patterns that represent characteristic shapes in the data, then measure how well each signal matches these patterns using mean squared error (MSE). These MSE values become features for prediction.

The key insight is that discriminative patterns—shapes that differ systematically between classes or correlate with outcomes—will produce MSE features that separate samples effectively. By searching for patterns across multiple signal representations (raw signals, derivatives, frequency spectra, wavelets), patX can discover features in whichever domain is most informative for a given task.

An overview of the patX methodology is shown in Figure 1.



Figure 1: patX methodology overview: The process begins with time series or spatial data, applies multiple signal transformations, and selects the top-K transforms via cross-validation. Iterative pattern discovery then: (1) proposes candidate pattern parameters (channel, transform, control points, position range, width), (2) generates a spline pattern, (3) computes min-window MSE dissimilarity features between pattern and transformed signal segments, and (4) evaluates model performance via cross-validation. The best pattern is appended to the feature set and the procedure repeats until adding a new pattern no longer improves performance.

3.2. Signal Transformation and Selection

The first stage applies a set of signal transformations to all input channels, creating multiple views of the data. The default transformation library includes time-domain (raw, delta-to-first sample, percent change, cumulative sum, difference), frequency-domain (DCT, FFT power), wavelet decom-

positions (Coiflet-1), and nonlinear transforms (log1p, absolute value, soft-sign, reciprocal, sorting, pre-emphasis). Users can define and register custom transformations for domain-specific applications.

To focus the pattern search on the most informative representations, patX evaluates each transformation independently. For each transform, it computes aggregate statistics (mean, standard deviation, skewness, kurtosis, quartiles, etc.) across all channels, then trains a model using cross-validation. The top-K transforms by validation score are retained for pattern optimization. This automatic selection ensures patterns are discovered in the signal representations most relevant to the prediction task.

3.3. Pattern Characteristics Optimization

The core of patX is an optimization loop that discovers discriminative patterns. Each pattern is defined by five parameters:

1. **Channel index:** which input channel to examine (for multivariate data)
2. **Transform type:** which of the top-K transforms to use
3. **Control points:** values defining the spline shape
4. **Search interval:** allowed start/end range for matching
5. **Width:** temporal extent of the pattern

patX represents each pattern as a quadratic spline (a quadratic B-spline / Bezier curve) parameterized by three control points [36]. This yields smooth, human-readable motifs while keeping the parameter space compact.

Figure 2 illustrates the pattern generation and matching mechanism. Panel A shows how a smooth quadratic B-spline template $P(t)$ is calculated from three learnable control points (c_0, c_1, c_2) using the quadratic Bezier formulation: $P(t) = (1 - t)^2 c_0 + 2(1 - t)t c_1 + t^2 c_2$, where $t \in [0, 1]$. These control points are optimized within the range $[-1, 1]$ relative to the signal’s local amplitude to discover task-specific shapes.

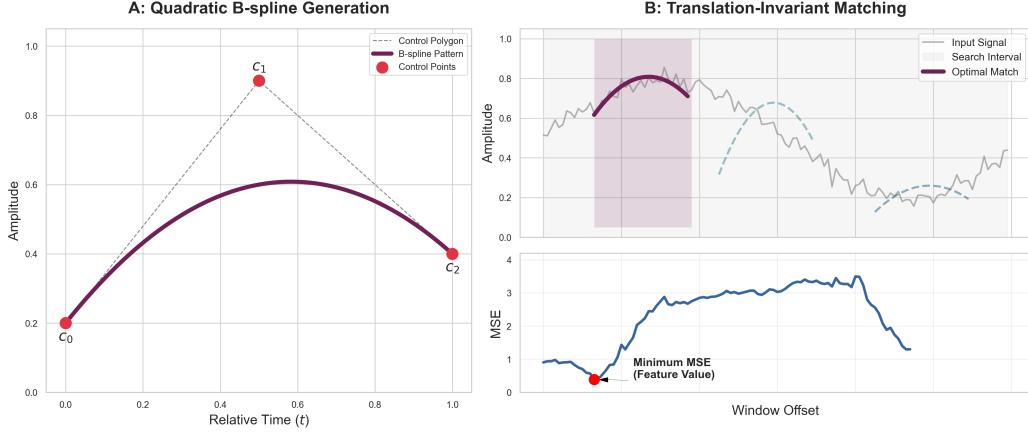


Figure 2: Illustration of pattern generation and matching. (A) A quadratic B-spline pattern is generated from three control points (c_0, c_1, c_2) defining the spline’s local curvature. (B) Translation-invariant matching: the pattern slides across the signal, generating a response map of dissimilarity scores (e.g., Mean Squared Error). The minimum value in this response map is selected as the representative feature for the sample, analogous to global min-pooling in convolutional networks.

The optimizer (Optuna with a TPE sampler) proposes candidate patterns. For each candidate, patX:

1. Generates the spline pattern from control points
2. Computes the dissimilarity between pattern and signal (default: MSE)

To support spatially invariant pattern detection, patX can evaluate each pattern in a sliding window over an allowed position range and use the best match (minimum MSE or maximum correlation) as the feature value (see Figure 2B-C). This is analogous to applying a convolutional filter across positions followed by a global pooling operation: the learned pattern is reused at every offset, and the pooling step makes the feature insensitive to small shifts. patX also supports Pearson correlation as a distance metric ($1 - \rho$), allowing for scale-invariant pattern discovery. If the chosen width is larger than the allowed search interval, patX falls back to evaluating the pattern at the interval center.

Each pattern contributes one MSE feature per sample. After optimizing one pattern, patX appends it to the current feature matrix and repeats the search for an additional pattern. The loop stops when the best new candidate fails to improve inner cross-validation performance.

3.4. Output

The final output consists of:

- A list of interpretable B-spline patterns with their parameters (channel, transform, control points, position, width)
- A trained machine learning model
- Feature matrices for train and test sets

Each pattern can be visualized by plotting its B-spline curve overlaid on example signal segments, enabling domain experts to validate whether discovered patterns correspond to known physiological or biological phenomena.

3.5. Implementation Details

Table 2 summarizes the default configuration used in our experiments. All parameters are configurable; patX supports alternative optimizers, machine learning models, B-spline configurations, and custom signal transformations. patX is available as an official Python package on PyPI (<https://pypi.org/project/patx/>).

Table 2: patX Default Configuration

Component	Default Setting
Optimizer	Optuna with TPE sampler
Trials per pattern	1,000
B-spline Degree	Quadratic (degree 2)
Control Points	3 per pattern
Transforms Library	13 built-in (time/frequency/wavelet/nonlinear)
Transforms Selected	Top-5 by CV performance
Inner CV Folds	4
ML Model	LightGBM

The built-in transformation library includes: raw, delta0, pct_change, cumsum, diff, log1p, abs, softsign, sorted, DCT, reciprocal, preemph99, FFT power, and a Coiflet-1 wavelet transform. Users can extend this library with domain-specific transformations. We ran our experiments on a MacBook Pro M2 using 12 cores and 32GB of RAM.

3.6. Datasets

We evaluate patX across seven diverse biomedical datasets spanning glucose forecasting, arrhythmia classification, emotion classification, gene expression prediction, clinical outcome prediction, human activity recognition, and voice pathology detection. Table 3 summarizes the key characteristics of each dataset.

Table 3: Dataset Characteristics and Evaluation Metrics

Dataset	Samples	Chan.	Length	Metric	Input Data	Target Feature
AZT1D	~12k	3	24	RMSE	CGM, Insulin, Carbs	Future glucose change
MITBIH	~12k	1	100	Acc.	ECG beats	5-class arrhythmia type
Emotions	2,048	5	254	AUC	EEG channels	3-class emotion state
REMC	18,421	5	200	AUC	Histone marks	Gene expression (high/low)
MIMIC-IV	~4k	29	24	AUC	EHR time series	ARDS diagnosis
PAMAP2	~4k	51	100	Acc.	IMU sensors	12-class activity type
SVD	~2k	3	700	AUC	Sustained vowels (/a/,/i/,/u/)	Healthy vs. pathology

AZT1D (glucose forecasting). Glucose change prediction from 24 lagged timepoints (2h history) of CGM, insulin, and carbohydrate signals, with physiological response curve modeling for delayed effects (see Supplementary Figure 6). Link: <https://data.mendeley.com/datasets/gk9m674wcx/1>.

MITBIH (arrhythmia classification). We extract 100-sample ECG beats centered on R-peaks, classifying into five arrhythmia types (N, L, R, A, V) with random undersampling for class balance. Link: <https://www.kaggle.com/datasets/mondejar/mitbih-database>.

Emotions (EEG emotion classification). EEG recordings (254 time points, 5 channels) during emotional states, classified into positive, neutral, and negative emotions. Link: Kaggle dataset.

REMC (ChIP-seq gene expression). Binary gene expression prediction from five histone marks (H3K4me1, H3K4me3, H3K36me3, H3K9me3, H3K27me3) across 200 bins around the TSS for 18,421 genes. Data: https://egg2.wustl.edu/roadmap/web_portal/processed_data.html.

MIMIC-IV (ARDS classification). ARDS prediction from 29 clinical time series (vitals, labs, ventilator settings) over 24 hourly bins per ICU patient [23]. Link: <https://physionet.org/content/mimiciv/3.1/>.

PAMAP2 (activity recognition). Activity classification from 51 IMU

channels (3 sensors $\times$ 17 features) using 100-sample windows across 12 activity classes from 9 subjects [24].

SVD (voice pathology detection). Binary classification of healthy versus pathological voice recordings from the Saarbrücken Voice Database (SVD), following the methodology of Lee [37]. We select 259 healthy and 259 pathological recordings per gender at normal pitch for sustained vowels /a/, /i/, and /u/. Female pathologies include hyperfunctional dysphonia (97), functional dysphonia (51), laryngitis (30), and others (81); male pathologies include laryngitis (62), hyperfunctional dysphonia (44), vocal fold polyp (25), and others (128). Raw audio waveforms are bandpass filtered (80–4000 Hz), pre-emphasized, silence-trimmed, and resampled to 700 samples per vowel. Following established protocols, we perform separate evaluations by gender and vowel to account for physiological differences in vocal characteristics [38, 39]. Link: <https://zenodo.org/records/16874898>.

3.7. Baselines

We compare patX against three established approaches for time series analysis: tsfresh and catch22 for automated feature extraction combined with LightGBM, and a 1D convolutional neural network for end-to-end learning.

3.7.1. TSFRESH and catch22 Baselines

For tsfresh, we use `MinimalFCParameters` to extract compact descriptors including moments, spectral features, and shape statistics, training LightGBM on all extracted features. For catch22, we extract the 22 canonical features from the library [9] and train LightGBM. Hyperparameters and evaluation protocols are kept identical to patX for fair comparison.

3.7.2. CNN Baseline Implementation

We implement a 1D CNN using PyTorch with three convolutional layers (filters: 64, 128, 256) each containing convolution, batch normalization, ReLU, max pooling, and dropout. The architecture uses kernel sizes of 7, 5, and 3 for the three layers respectively, with appropriate padding to maintain temporal resolution. Adaptive average pooling aggregates features across the temporal dimension, followed by fully connected layers (256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ output) with ReLU activations, batch normalization, and dropout (0.4, 0.3) for regularization. We train with Adam optimizer (lr=0.1, weight decay=1e-4), early stopping (patience=10), and learning rate reduction on plateau (factor=0.5, patience=3). For binary classification tasks, we use cross-entropy

loss with class weights computed to handle imbalanced datasets; for multi-class classification and regression, we use unweighted cross-entropy and MSE loss respectively. The model automatically detects the number of input channels based on the dataset structure: 3 channels for AZT1D (CGM, insulin, carbs), 5 channels for REMC (histone marks), and 1 channel for univariate time series. Training uses 100 epochs maximum with batch size 128, matching evaluation protocols with other baselines.

3.8. *Evaluation*

For datasets using cross-validation, we use 5-fold stratified splits with fixed seeds. For subject-level datasets (AZT1D, PAMAP2), we use 70%/30% train/test splits per subject. Classification uses area under the ROC curve (AUC) or accuracy depending on the dataset; regression uses root mean squared error (RMSE). We report the number of extracted patterns alongside predictive performance as mean $\pm$ standard deviation across folds or subjects.

4. Results

4.1. Performance Comparison

Table 4: Performance comparison across biomedical datasets. Results show mean $\pm$ standard deviation across subjects/folds. Best performance per dataset highlighted in bold.

Dataset	Method	Score	Features	Time (s)
AZT1D	PATX	24.482 ± 3.796	14.0 ± 3.6	472.1 ± 126.1
	TSFRESH	32.546 ± 4.961	11.0	1.7 ± 0.2
	CATCH22	32.342 ± 4.808	23.0	2.8 ± 0.4
	CNN	30.176 ± 4.494	73.0	21.3 ± 11.4
Emotions	PATX	0.979 ± 0.006	7.6 ± 1.3	356.0 ± 49.4
	TSFRESH	0.904 ± 0.008	10.0	1.0 ± 0.0
	CATCH22	0.876 ± 0.005	22.0	12.4 ± 0.2
	CNN	0.668 ± 0.001	1500.0	3.3 ± 1.1
MIMIC-IV	PATX	0.788 ± 0.007	10.0 ± 3.5	293.8 ± 100.4
	TSFRESH	0.738 ± 0.007	10.0	1.2 ± 0.0
	CATCH22	0.752 ± 0.003	22.0	8.6 ± 0.1
	CNN	0.776 ± 0.007	696.0	3.4 ± 0.5
MITBIH	PATX	0.947 ± 0.004	14.6 ± 2.2	1274.6 ± 218.2
	TSFRESH	0.527 ± 0.008	10.0	2.1 ± 0.0
	CATCH22	0.710 ± 0.013	22.0	4.0 ± 0.0
	CNN	0.939 ± 0.004	100.0	28.9 ± 3.6
REMC	PATX	0.931 ± 0.008	14.5 ± 3.2	631.0 ± 141.6
	TSFRESH	0.871 ± 0.018	10.0	6.0 ± 0.1
	CATCH22	0.897 ± 0.009	22.0	53.4 ± 1.4
	CNN	0.932 ± 0.008	1000.0	14.9 ± 4.1
PAMAP2	PATX	0.997 ± 0.001	9.8 ± 1.3	3895.2 ± 562.9
	TSFRESH	0.465 ± 0.005	10.0	9.5 ± 0.1
	CATCH22	0.484 ± 0.005	22.0	80.0 ± 3.2
	CNN	0.985 ± 0.008	520.0	242.6 ± 39.7
SVD	PATX	0.616 ± 0.009	5.2 ± 1.3	191.5 ± 42.7
	TSFRESH	0.576 ± 0.011	10.0	2.1 ± 0.1
	CATCH22	0.570 ± 0.015	22.0	51.8 ± 0.6
	CNN	0.571 ± 0.003	6300.0	1.0 ± 0.5

Table 4 presents the test set performance of patX + LightGBM, ts-fresh + LightGBM, catch22 + LightGBM, and CNN baselines across seven biomedical datasets, now including the Saarbrücken Voice Database (SVD) for healthy vs. diseased voice pathology detection.

patX achieves the best overall performance on six of seven datasets (AZT1D, Emotions, MIMIC-IV, MITBIH, PAMAP2, and SVD). On REMC, the CNN baseline slightly outperforms patX (0.932 vs. 0.931). The gains are especially pronounced on AZT1D glucose forecasting, where patX reduces RMSE from 30.2 (CNN) and 32.5 (tsfresh) to 24.5 while using about 14 patterns on average.

We focus on two aspects: predictive performance and feature transparency. patX produces a compact set of explicit patterns and turns their matches into features, enabling direct visualization and domain validation. As a result, the model’s decision inputs can be inspected in the same units as the original signals (or their selected transforms).

4.2. Ablation Study

We ablate key components across six datasets using identical 5-fold stratified splits and gradient boosting classifiers. Table 5 reports mean $\pm$ std for score, runtime, and feature count for each variant: default configuration, reduced inner CV (1 vs. 4 folds), reduced trials (100 vs. 1000), raw-only transforms, subsampling (50% of training data), Pearson correlation as distance metric, and linear regression as the downstream ML model.

Table 5: Ablation study across six datasets. Bold indicates best score per dataset. Arrows indicate optimization direction ($\uparrow$ higher better, $\downarrow$ lower better).

Dataset	Variant	Score	Time (s)	Features
MITBIH (Accuracy $\uparrow$)	Default	0.947 ± 0.003	615 ± 112	12 ± 3
	Inner CV=1	0.946 ± 0.003	207 ± 48	12 ± 2
	No early stopping	0.949 ± 0.005	1956 ± 362	14 ± 3
	100 trials	0.948 ± 0.002	254 ± 57	16 ± 3
	No transforms	0.946 ± 0.005	1107 ± 221	16 ± 3
	No shifting	0.900 ± 0.018	1130 ± 264	14 ± 2
	1 control point	0.931 ± 0.004	939 ± 282	15 ± 4
SVD (Accuracy $\uparrow$)	Default	0.621 ± 0.011	48 ± 17	5 ± 2
	Inner CV=1	0.612 ± 0.015	24 ± 5	4 ± 1
	No early stopping	0.614 ± 0.007	181 ± 66	6 ± 3
	100 trials	0.630 ± 0.032	16 ± 6	7 ± 3
	No transforms	0.594 ± 0.027	48 ± 18	5 ± 2
	No shifting	0.568 ± 0.048	41 ± 8	5 ± 1
	1 control point	0.611 ± 0.011	60 ± 28	5 ± 2
REMC (E003) (AUC $\uparrow$)	Default	0.930 ± 0.004	386 ± 54	17 ± 1
	Inner CV=1	0.927 ± 0.005	146 ± 18	12 ± 1
	No early stopping	0.931 ± 0.004	1023 ± 185	18 ± 3
	100 trials	0.930 ± 0.005	106 ± 31	19 ± 5
	No transforms	0.928 ± 0.004	386 ± 59	16 ± 2
	No shifting	0.928 ± 0.004	315 ± 35	13 ± 1
	1 control point	0.930 ± 0.006	303 ± 55	15 ± 2
Emotions (Accuracy $\uparrow$)	Default	0.966 ± 0.030	133 ± 70	9 ± 5
	Inner CV=1	0.960 ± 0.009	42 ± 10	6 ± 2
	No early stopping	0.958 ± 0.025	370 ± 130	9 ± 3
	100 trials	0.959 ± 0.024	30 ± 12	9 ± 3
	No transforms	0.923 ± 0.014	109 ± 25	7 ± 2
	No shifting	0.912 ± 0.020	121 ± 27	9 ± 2
	1 control point	0.964 ± 0.022	135 ± 24	9 ± 2
MIMIC (AUC $\uparrow$)	Default	0.778 ± 0.015	81 ± 35	8 ± 4
	Inner CV=1	0.764 ± 0.017	24 ± 5	5 ± 2
	No early stopping	0.777 ± 0.019	295 ± 58	11 ± 2
	100 trials	0.769 ± 0.016	21 ± 2	7 ± 1
	No transforms	0.765 ± 0.011	80 ± 33	9 ± 4
	No shifting	0.779 ± 0.011	95 ± 37	10 ± 4
	1 control point	0.789 ± 0.006	86 ± 14	10 ± 1
AZT1D (RMSE $\downarrow$)	Default	12.29 ± 0.65	1115 ± 208	15 ± 3
	Inner CV=1	13.09 ± 0.55	234 ± 49	12 ± 3
	No early stopping	12.10 ± 0.31	3386 ± 1067	19 ± 5
	100 trials	13.22 ± 0.93	327 ± 61	19 ± 4
	No transforms	10.03 ± 0.37	2390 ± 287	25 ± 2
	No shifting	20.70 ± 1.17	337 ± 107	12 ± 2
	1 control point	11.59 ± 0.28	1727 ± 250	23 ± 2

The ablations reveal dataset-dependent component importance:

Sliding-window matching is critical for temporally misaligned data. On MITBIH, disabling sliding-window matching causes the largest drop (0.900 vs. 0.947), confirming that translation-invariant matching is essential for aligning ECG beat morphology across patients. Similarly, on AZT1D glucose forecasting, disabling sliding-window matching degrades RMSE dramatically (20.70 vs. 12.29), indicating that temporal alignment of glucose dynamics varies substantially across prediction windows.

Transforms provide task-specific gains. On AZT1D, disabling transforms (raw-only) actually *improves* performance (10.03 vs. 12.29 RMSE), suggesting that glucose patterns are best captured directly in the time domain without spectral or wavelet views. Conversely, on Emotions EEG data, no transforms hurts substantially (0.923 vs. 0.966), indicating that frequency-domain representations are crucial for emotion classification.

Optimization budget shows diminishing returns. On REMC, 100 trials matches the default (0.930 AUC) at $3.6\times$ lower runtime. On SVD, 100 trials actually achieves the best score (0.630 vs. 0.621), suggesting that excessive optimization can overfit on smaller datasets. On MITBIH, increasing the search budget yields only marginal gains (0.949 vs. 0.947) but at substantially higher runtime cost.

Inner CV reduction trades accuracy for speed. Reducing inner CV from 4 to 1 fold cuts runtime 2–4 $\times$ with modest score drops (e.g., 0.946 vs. 0.947 on MITBIH, 0.764 vs. 0.778 on MIMIC).

Flexibility in distance metrics and models. The addition of Pearson correlation as a distance metric and Linear Regression as a downstream model demonstrates patX’s modularity. Pearson distance can be more robust to amplitude scaling, while linear models offer the highest level of transparency for interpreting pattern contributions.

Subsampling enables efficient search on large datasets. By optimizing patterns on a representative subset of the data (e.g., 50%), patX achieves substantial speedups during the search phase while maintaining competitive performance when the final model is trained on the full dataset.

4.3. Domain-Specific Pattern Interpretation

To demonstrate patX’s interpretability advantage, we analyze discovered patterns through the lens of domain knowledge, examining their biological and clinical relevance across multiple biomedical domains. The multi-representation search strategy enables patterns to be discovered in the most

discriminative signal space—whether time domain, frequency domain (via wavelets or FFT), or derivative space—enhancing both interpretability and predictive power.

Figure 3 presents a comprehensive visualization of pattern discovery in the REMC dataset, focusing on the H3K4me3 histone mark in cell line E003. The analysis is decomposed into four panels to illustrate the complete feature extraction process: (a) Raw input signals for representative high and low expression genes, showing H3K4me3 enrichment around the transcription start site (TSS). (b) The transformed signal space (cumulative sum) where pattern matching occurs, with the learned B-spline pattern overlaid at its optimal position. The cumsum transformation captures the total accumulation of histone marks, which is significantly higher for active promoters. (c) Average class profiles (mean $\pm$ std) demonstrating the global difference in cumulative H3K4me3 signal between high and low expression genes, with the pattern targeting the region of maximal divergence. (d) The distribution of MSE values, showing clear separation between classes based on their similarity to the learned pattern. This multi-view visualization confirms that patX discovers biologically meaningful patterns (H3K4me3 accumulation) and uses them to construct discriminative features in an interpretable manner.

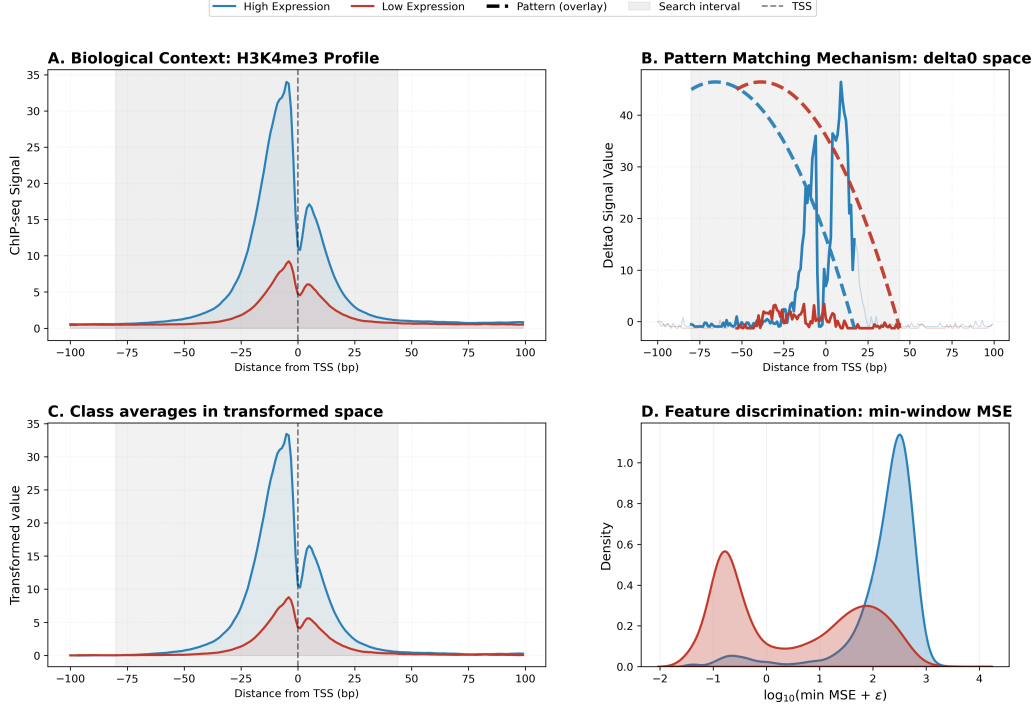


Figure 3: Domain-specific pattern interpretation for REMC (H3K4me3). The four panels illustrate the pattern extraction mechanism: (Top-Left) Raw ChIP-seq signals for representative high (blue) and low (red) expression genes around the TSS. (Top-Right) Transformed signals (cumulative sum) with the learned B-spline pattern (green) overlaid at its optimal matching position. (Bottom-Left) Average profiles for high and low expression classes with standard deviation shading, showing how the pattern aligns with the global difference in H3K4me3 accumulation. (Bottom-Right) Distribution of MSE features, demonstrating discriminative separation between gene expression classes based on pattern similarity.

4.3.1. Individual Prediction Explainability via SHAP Analysis

To demonstrate patX’s explainability at the individual prediction level, we perform SHAP (SHapley Additive exPlanations) analysis [40, 41] on a representative high-expression gene from REMC (E003). Figure 4 shows how discovered histone patterns combine to predict gene expression.

The resulting attributions can often be related to known biology. For example, patterns linked to promoter-associated marks (e.g., H3K4me3) tend to support predictions of active genes, while patterns linked to repressive marks (e.g., H3K27me3) tend to support the opposite direction [42, 43]. The automatic selection of both raw (profile shape) and cumulative transforma-

tions illustrates how patX can capture complementary signal aspects without manual specification.

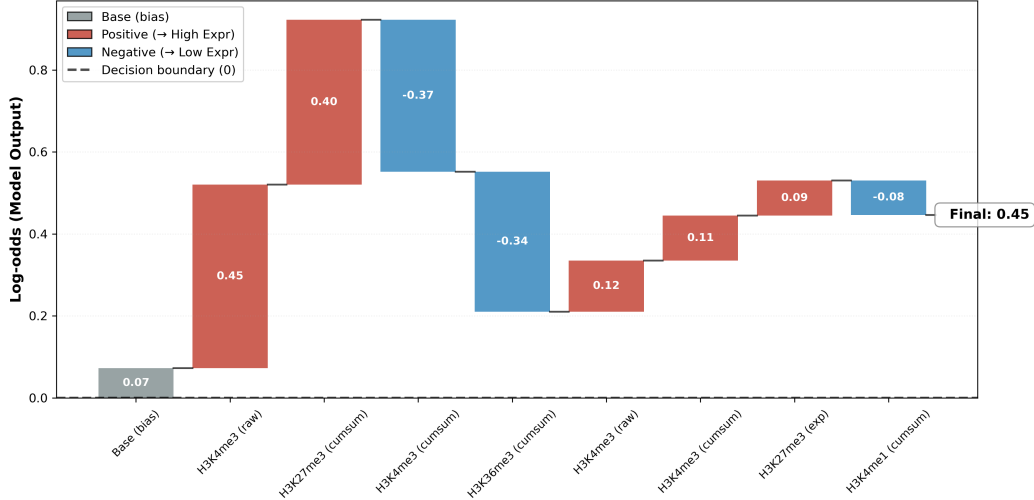


Figure 4: SHAP waterfall plot for gene expression prediction in REMC (E003). Each bar shows a pattern’s contribution to the final log-odds prediction. Feature labels indicate the histone mark and transformation type. Positive contributions (red) push towards high expression; negative (blue) towards low expression.

4.4. Pattern Stability and Reproducibility

To evaluate patX’s robustness, we analyze pattern consistency across cross-validation folds and subjects. Table 6 reports the number of selected patterns (mean $\pm$ std) and the dominant transformation and channel of the most important pattern with consistency percentages. Single-channel datasets (MITBIH, Emotions) show “—” for channel consistency.

Across datasets, we observe that some tasks favor a small set of dominant transformations or channels, while others show more diversity. This aligns with differences in signal structure: some problems have a clear dominant representation (e.g., rate-of-change or frequency content), whereas others distribute discriminative information across channels and transforms.

Figure 5 visualizes the primary pattern from each fold for REMC (E003), showing consistent positioning near the transcription start site despite variation in histone mark selection.

Table 6: Pattern discovery characteristics across datasets. For each dataset, we report the number of selected patterns (mean $\pm$ std across folds/subjects), and the dominant transformation and channel of the most important pattern with consistency percentages. Single-channel datasets show “—” for channel consistency.

Dataset	Patterns	Transform (consistency)	Channel (consistency)
MITBIH	11.8 ± 2.7	pct_change (60%)	ECG (—)
EMOTIONS	5.6 ± 1.5	preemph97 (60%)	EEG (60%)
MIMIC	6.0 ± 1.9	dct (100%)	pao2 (40%)
PAMAP2	8.4 ± 1.6	sorted (40%)	heart_rate (20%)
SVD	2.4 ± 1.5	ecg_resp (20%)	a_n_mfcc1 (60%)
REMC (E003)	11.6 ± 1.4	delta0 (40%)	H3K4me3 (100%)
AZT1D	6.4 ± 1.3	raw (29%)	CGM (62%)

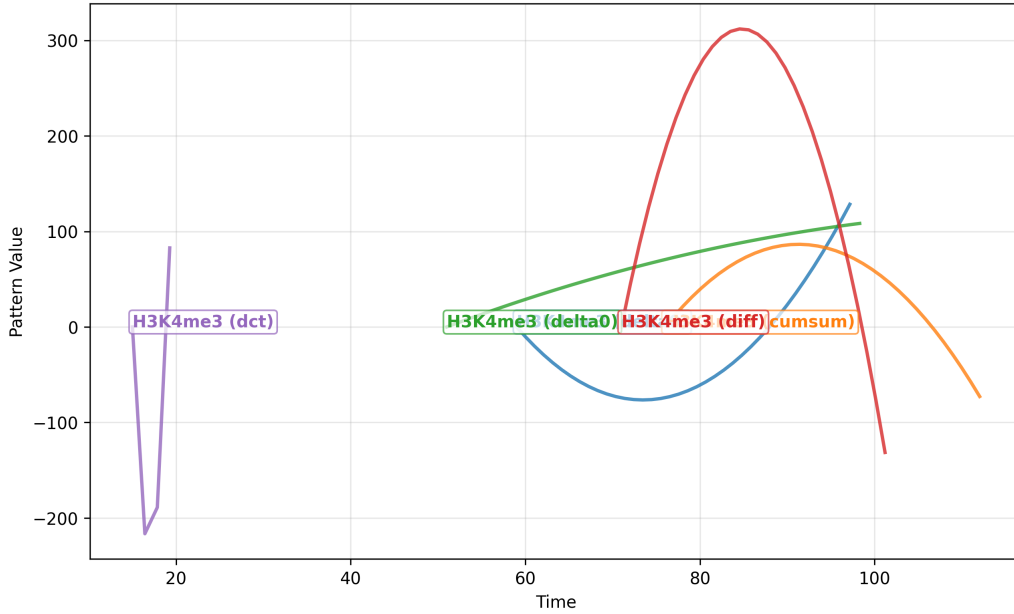


Figure 5: Pattern stability analysis for REMC (E003). Each colored curve represents the most important B-spline pattern discovered in one cross-validation fold, with annotations indicating the histone mark and transformation type. Patterns consistently target the region near the transcription start site, demonstrating positional stability across data splits.

5. Discussion

Biomedical deployment places additional constraints on learning systems beyond benchmark accuracy: clinicians and scientists must be able to *inspect* what evidence a model uses, relate it to domain knowledge, and understand when and why failures occur [4, 35]. In this work, patX operationalizes a practical middle ground between hand-engineered descriptors and opaque end-to-end representations by turning pattern discovery into explicit, human-readable features.

5.1. *Why patterns are a useful interface in biomedicine*

patX represents each sample through its dissimilarity to a small set of learned patterns. This pattern-based interface is attractive for biomedical use for three reasons. First, patterns are visual: they can be overlaid on signals and interpreted in units that domain experts already reason about (e.g., ECG morphology, spectral structure in EEG, accumulation profiles in epigenomics). Second, patterns are local: even when matching is translation-invariant via sliding-window min-MSE pooling, the method still yields a concrete location of the best match, supporting case-by-case review. Third, patterns provide a natural bridge to earlier interpretable subsequence literature such as shapelets and motif discovery, which have long emphasized compact, representative subsequences as explanatory evidence [13, 15, 17, 18].

5.2. *Efficiency and deployability*

patX is designed around a simple computation graph: precompute a small set of transformations, match a handful of patterns, and run a lightweight gradient-boosting model on the resulting feature matrix. This structure is appealing in biomedical pipelines where compute budgets and reproducibility constraints are common, and where CPU-based inference is often preferred in production. Unlike large automated feature sets, the learned representation is compact by construction, and unlike deep models, it does not require specialized hardware or complex runtime dependencies for inference.

The ablation study (Table 5) clarified which patX components matter across domains. On MITBIH and AZT1D, disabling sliding-window matching causes the largest performance drops, supporting the need for translation-invariant matching when temporal alignment varies across samples. Conversely, on REMC where signals are anchored to the TSS, sliding provides

no benefit. The transform library shows task-specific value: raw-only improves AZT1D glucose forecasting while hurting Emotions EEG classification, suggesting that domain-appropriate signal representations matter more than exhaustive transform coverage. Optimization budget shows diminishing returns: 100 trials matches or exceeds default performance on REMC and SVD at 3–4 $\times$ lower runtime. Overall, these results suggest that sliding-window matching, transforms, and search budget should be treated as dataset-dependent knobs rather than fixed defaults.

Interpretability in patX comes from two sources: (i) the explicit pattern shape and its parameters, and (ii) the localization of where the pattern matches the signal. This enables domain experts to validate whether extracted patterns align with known physiology (e.g., ECG wave components, histone-mark enrichment near the TSS) and to inspect failure modes by examining which patterns activate. In addition, feature attribution methods can be applied directly to the learned pattern features, yielding explanations in the same units as the matching score.

5.3. Limitations and future work

patX trades end-to-end flexibility for transparency. It relies on a predefined transform library and a pattern family, which may miss rare or highly irregular motifs. Runtime can be substantial when increasing the optimization budget, and the best settings can vary across datasets, motivating automated configuration selection. Future work includes learned or domain-specific transforms, more efficient search strategies, richer pattern parameterizations, and interactive workflows that pair pattern discovery with expert feedback.

6. Conclusion

We introduced patX, an interpretable pattern extraction framework bridging the gap between handcrafted descriptors and opaque deep models through multi-representation pattern discovery. Through extensive experiments across seven diverse biomedical datasets, we demonstrated that a small set of automatically discovered B-spline motifs can provide transparent, task-specific features for standard machine learning models, while keeping the learned representations human-readable and directly visualizable. The optional sliding-window matching enables translation-invariant pattern detection, analogous to applying a convolutional filter followed by pooling, while retaining explicit

pattern parameters. The B-spline representation combined with automatic transformation selection provides smooth, locally controllable patterns that capture both shape and amplitude information across multiple signal representations (time, frequency, derivative domains). Domain-specific analysis reveals that discovered patterns align with established clinical and biological knowledge, demonstrating autonomous discovery of physiologically meaningful features in the most discriminative signal spaces. Future extensions will explore adaptive control point allocation, learned or domain-specific signal transformations, semi-supervised pretraining on large unlabeled corpora, and tighter integration with interactive visualization tools to facilitate clinical adoption.

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7. Supplementary Data

7.1. Signal transformations and usage

patX builds a fixed library of transformations and lets the optimizer pick the most predictive representations. The library spans time, frequency, and

nonlinear views: raw, delta0, pct_change, cumulative sum, difference, log1p, absolute value, softsign, sorted, DCT, reciprocal, pre-emphasis, FFT power, and a Coiflet-1 wavelet transform. Each transform is applied per channel to produce a parallel stack of views of the same signals.

During training, patX first scores every transform independently using aggregate statistics (mean, std, quantiles, skew, kurtosis, range, halves of the series) and cross-validation, then retains the top-K transforms for pattern search. For every candidate pattern, the optimizer selects both a transform and channel, fits a spline pattern, and measures MSE-based dissimilarity in the chosen transformed space. The final model only computes features for the selected transforms, ensuring consistency between training and inference while keeping the search broad but the deployed pipeline compact.

7.2. Dataset Processing Details

7.2.1. AZT1D Preprocessing: Insulin and Carbohydrate Response Modeling

The AZT1D dataset presents unique challenges for glucose forecasting due to the complex temporal dynamics of insulin and carbohydrate effects on blood glucose levels. Unlike simple time series where immediate effects are observed, physiological responses involve significant delays: insulin takes 10–90 minutes to affect glucose levels, while carbohydrates impact glucose within 10–60 minutes, with effects persisting for 2–4 hours [25].

To address these delayed physiological responses, we implement knowledge-driven response curve modeling based on established pharmacokinetic and glycemic response literature. For insulin, we use a gamma-like function with 15-minute onset, 60-minute peak, and 4-hour duration; for carbohydrates, we use 10-minute onset, 45-minute peak, and 3-hour duration. These response curves model the temporal dynamics of physiological effects, with insulin effects peaking around 60 minutes and persisting for up to 4 hours, while carbohydrate effects peak around 45 minutes and persist for up to 3 hours.

At each timepoint, we compute cumulative active effects by convolving all historical insulin doses and carbohydrate intakes with their respective response curves. This creates interpretable time series that capture overlapping physiological effects, enabling the model to understand how multiple insulin doses and meal intakes interact over time.

The preprocessing pipeline extracts 24 timepoints (2 hours of history at 5-minute intervals) for three physiological signals: CGM values, active insulin effects, and active carbohydrate effects. The regression target is future glucose change $g_{t+h} - g_t$ where $h = 30$ minutes (6 timepoints ahead). We

also include the initial CGM value (CGM.0) as a baseline feature to provide current glucose context.

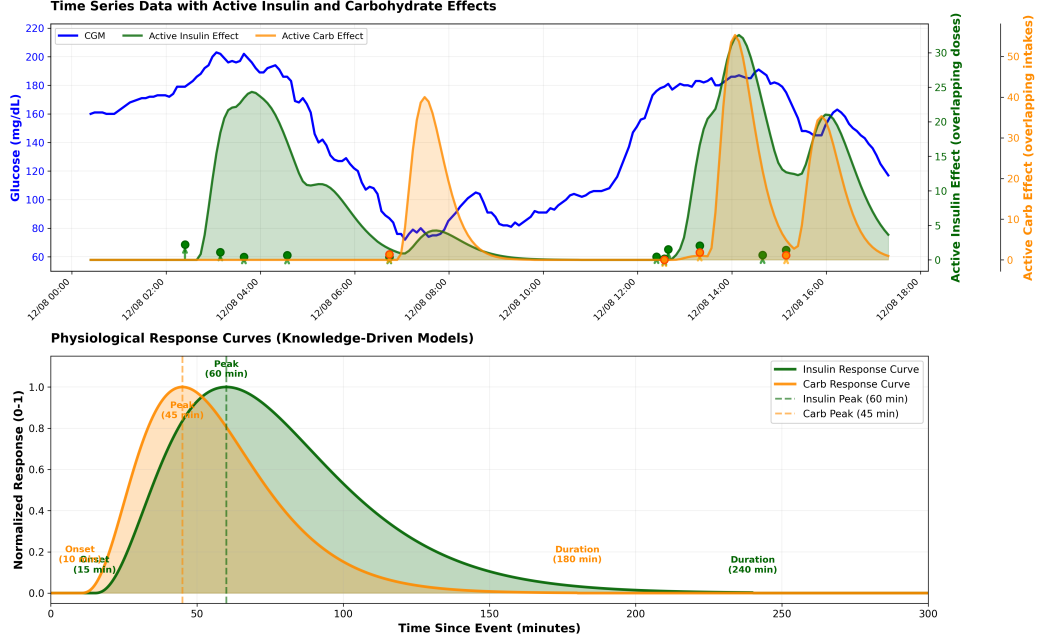


Figure 6: AZT1D preprocessing visualization showing insulin and carbohydrate response modeling. Top panel: Real-time CGM with overlapping insulin and carbohydrate effects over time for Subject 1. Blue line shows CGM values with target range (70-180 mg/dL) highlighted in green. Dark green shaded area represents active insulin effect from overlapping doses (circles with crosses). Orange shaded area represents active carbohydrate effect from overlapping intakes (circles with crosses). Bottom panel: Physiological response curves showing normalized insulin (15-min onset, 60-min peak, 240-min duration) and carbohydrate (10-min onset, 45-min peak, 180-min duration) effects over time. These knowledge-driven models enable patX to learn from physiologically meaningful features that capture delayed and overlapping effects.

This preprocessing approach transforms raw event data into continuous physiological signals that capture the complex temporal dynamics of glucose regulation, providing patX with meaningful features for pattern discovery across multiple signal representations.

7.2.2. MITBIH Preprocessing

The MIT-BIH Arrhythmia Database contains annotated ECG recordings from 48 patients. We extract individual heartbeats by centering 100-sample

windows around annotated R-peak positions, with 50 samples before and 50 samples after each R-peak. Each beat is classified into one of five arrhythmia types: Normal (N), Left bundle branch block (L), Right bundle branch block (R), Atrial premature (A), and Premature ventricular contraction (V). To address class imbalance, we apply random undersampling to balance class distributions before training.

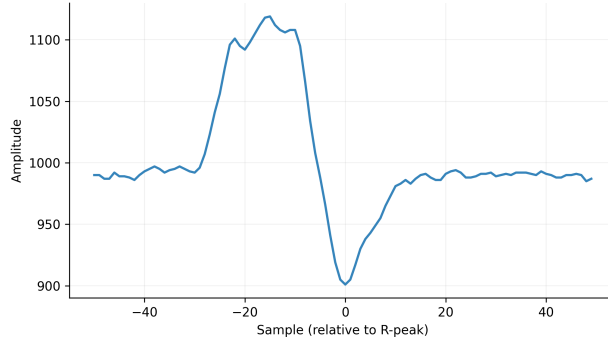


Figure 7: Representative ECG beat from MITBIH dataset showing characteristic R-peak morphology.

7.2.3. Emotions Preprocessing

The EEG emotions dataset contains brainwave recordings collected while participants experienced different emotional states (positive, neutral, negative). We extract FFT-derived features from 5 EEG channels (AF3, F7, F3, FC5, T7), yielding 254 frequency bins per channel. The FFT features are organized into two groups (a and b suffixes) representing different frequency decomposition components, providing 5 input channels with 254 time points each for pattern discovery.

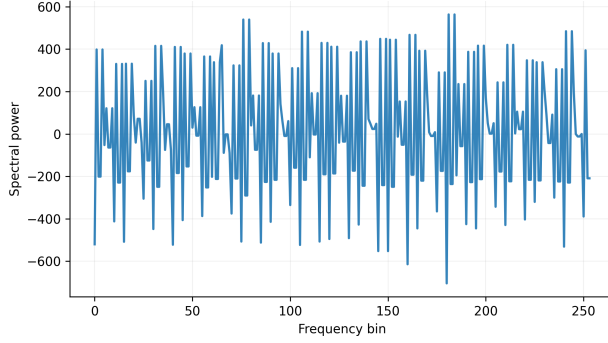


Figure 8: Representative EEG frequency spectra from Emotions dataset showing power distributions across five channels.

7.2.4. REMC Preprocessing

Following our prior work on PatternChrome [44], we predict binary gene expression (high/low) from histone modification ChIP-seq profiles from the Roadmap Epigenomics Consortium (56 cell lines, 19,795 genes). We convert RefSeq to GENCODE identifiers, retaining 18,421 genes. For each gene, we extract five histone modification signals (H3K4me1, H3K4me3, H3K36me3, H3K9me3, H3K27me3) from 5000 bp upstream and downstream of the TSS, divided into 50 bp bins (200 bins per mark). Gene expression is binarized at the median level. We split 6,600/5,911/5,910 genes for train/validation/test with balanced labels.

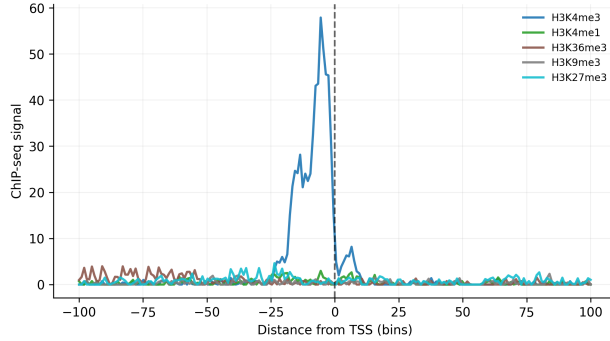


Figure 9: Representative histone modification profiles from REMC dataset showing all five marks (H3K4me3, H3K4me1, H3K36me3, H3K9me3, H3K27me3) around the transcription start site for a high-expression gene.

7.2.5. MIMIC-IV Preprocessing

We predict ARDS from multivariate clinical time series in ICU patients. The dataset contains 29 features per subject: vital signs (respiratory rate, heart rate, O2 saturation, blood pressure, temperature), ventilator settings (FiO2, PEEP), Glasgow Coma Scale scores, laboratory values (arterial pH, PaO2, creatinine, platelets, WBC, INR, lactate, bilirubin), vasopressors (norepinephrine, phenylephrine, epinephrine, dopamine, dobutamine), urine output, and derived features (PaO2/FiO2 ratio, pulse pressure). We extract 24-hour time series binned hourly, with forward/backward filling for missing values and 0.1/99.9th percentile capping for extreme values. Subjects with <5 hours of data are excluded; anchor_age is provided as baseline.

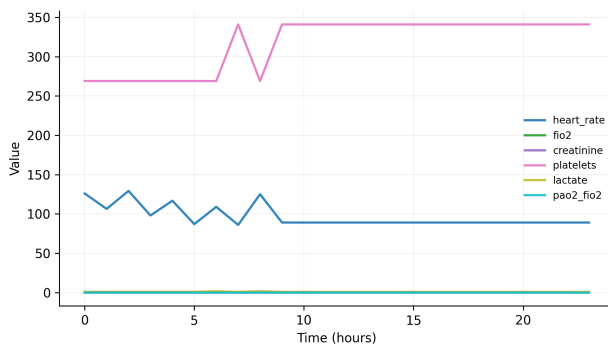


Figure 10: Representative clinical time series from MIMIC-IV dataset showing 24-hour trajectories of vital signs, laboratory values, and ventilator settings.

7.2.6. PAMAP2 Preprocessing

The PAMAP2 dataset contains recordings from 9 subjects performing 18 physical activities while wearing 3 IMUs (wrist, chest, ankle) and a heart rate monitor. Each IMU records temperature, 3D acceleration ($\pm 16g$ and $\pm 6g$ scales), 3D gyroscope, 3D magnetometer, and 4D orientation at 100Hz, yielding 51 channels. We create 100-sample windows (1 second) with 50% overlap, filtering transient activities (ID 0) to obtain 12 activity classes: lying, sitting, standing, walking, running, cycling, Nordic walking, watching TV, computer work, ascending/descending stairs, and vacuum cleaning. Missing values are interpolated and filled with column means.

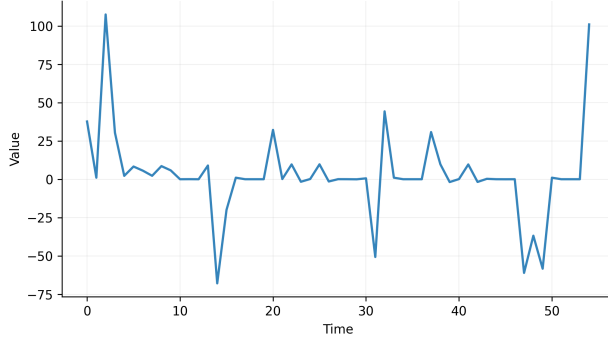


Figure 11: Representative IMU sensor data from PAMAP2 dataset showing acceleration and gyroscope measurements from hand, chest, and ankle sensors.

7.2.7. SVD Preprocessing

The Saarbrücken Voice Database (SVD) contains voice recordings from over 2000 subjects, including healthy controls and patients with 71 different voice pathologies [37]. Following the exact methodology of Lee [37], we select balanced subsets for male and female subjects separately:

- **Female:** 259 healthy + 259 pathological (97 hyperfunctional dysphonia, 51 functional dysphonia, 30 laryngitis, 81 other)
- **Male:** 259 healthy + 259 pathological (62 laryngitis, 44 hyperfunctional dysphonia, 25 vocal fold polyp, 128 other)

We extract sustained vowel recordings of /a/, /i/, and /u/ at normal pitch and apply the following preprocessing pipeline: (1) bandpass filtering (80–4000 Hz) to remove DC offset and high-frequency noise, (2) pre-emphasis with coefficient 0.97 to amplify higher frequencies, (3) silence trimming using RMS-based voice activity detection, (4) truncation to 3 seconds maximum, and (5) resampling to 700 samples using polyphase filtering. Each waveform is normalized to zero mean and unit variance, then clipped to ± 5 standard deviations.

Following Lee [37], we perform separate evaluations by gender and vowel to account for physiological differences in fundamental frequency and vocal tract characteristics between men and women.

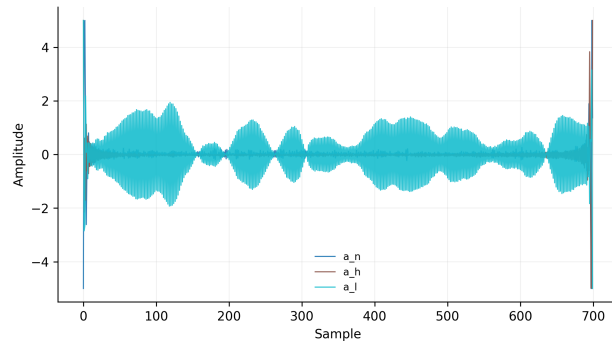


Figure 12: Representative waveforms from SVD dataset showing preprocessed audio signals for sustained vowels /a/, /i/, and /u/ at normal pitch. Each vowel is resampled to 700 samples after bandpass filtering and silence trimming.